

Ali Siahkoohi

Assistant Professor
Department of Computer Science
University of Central Florida

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<https://alishahkoohi.github.io>
Last updated: August 15, 2026

Research Interests

My research focuses on uncertainty quantification in complex systems that arise in computational science and engineering, with an emphasis on deep generative modeling. By integrating deep learning and scientific computing, my group develops scalable sampling techniques for high-dimensional distributions. This work enables Bayesian inference in large-scale PDE-based inverse problems (e.g., seismic and medical imaging) and provides a principled framework for building reliable, uncertainty-aware AI models for real-world applications.

Academic Appointments

University of Central Florida
Tenure-Track Assistant Professor
Department of Computer Science

August 2025 – Present
Orlando, FL, USA

Rice University
Simons Postdoctoral Fellow
Department of Computational Applied Mathematics & Operations Research
Jointly hosted by Maarten V. de Hoop and Richard G. Baraniuk

August 2022 – July 2025
Houston, TX, USA

Education

Georgia Institute of Technology
Ph.D. in Computational Science and Engineering
Advised by Felix J. Herrmann

August 2022
Atlanta, GA, USA

University of Tehran
M.Sc. in Geophysics

March 2016
Tehran, Iran

Sharif University of Technology
B.Sc. in Electrical Engineering

August 2013
Tehran, Iran

Publications

Google Scholar profile: <https://scholar.google.com/citations?user=sxRMqYIAAAAJ&h>

In Preparation & Under Review

- P5. [A. Siahkoohi](#) and S. Alemohammad. Priors learned from legacy reconstructions inherit undetectable overconfidence. Preprint arXiv:2607.21721, 2026
[pdf] [code] [bib]
- P4. [A. Siahkoohi](#). Amortized mean-shift interacting particles. Preprint arXiv:2606.15871, 2026
[pdf] [code] [bib]
- P3. [A. Siahkoohi](#) and A. Thatipelli. A lift for input-convex neural network training. Preprint arXiv:2605.24274, 2026
[pdf] [code] [bib]
- P2. [A. Siahkoohi](#), K. Aghazade, and A. Gholami. Dual-space posterior sampling for Bayesian inference in constrained inverse problems. Preprint arXiv:2603.00393, 2026a
[pdf] [code] [poster] [bib]

- P1. A. Thatipelli and [A. Siahkoohi](#). Hypernetwork-based approach for grid-independent functional data clustering. Preprint arXiv:2602.22823, 2026
[pdf] [code] [poster] [bib]

Journal Publications

- J10. A. Thatipelli, J. Sam, M. Louboutin, [A. Siahkoohi](#), R. Wang, and F. J. Herrmann. XConv: Low-memory stochastic backpropagation for convolutional layers. *Transactions on Machine Learning Research*, 2026
[pdf] [code] [link] [bib]
- J9. [A. Siahkoohi](#), R. Morel, R. Balestrieri, E. Allys, G. Sainton, T. Kawamura, and M. V. de Hoop. Multiscale clustering and source separation of InSight mission seismic data. *IEEE Transactions on Geoscience and Remote Sensing*, 64:1–16, 2026b
[pdf] [code] [slides] [link] [bib]
- J8. R. Orozco, [A. Siahkoohi](#), M. Louboutin, and F. J. Herrmann. ASPIRE: Iterative amortized posterior inference for Bayesian inverse problems. *Inverse Problems*, 41(4):045001, 2025
[pdf] [code] [link] [bib]
- J7. R. Orozco, P. Witte, M. Louboutin, [A. Siahkoohi](#), G. Rizzuti, B. Peters, and F. J. Herrmann. InvertibleNetworks.jl: A Julia package for scalable normalizing flows. *Journal of Open Source Software*, 9(99):6554, 2024
[pdf] [code] [link] [bib]
- J6. L. Luzzi, P. M. Mayer, J. Casco-Rodriguez, [A. Siahkoohi](#), and R. G. Baraniuk. Boomerang: Local sampling on image manifolds using diffusion models. *Transactions on Machine Learning Research*, 2024a
[pdf] [code] [link] [bib]
- J5. M. Louboutin, Z. Yin, R. Orozco, T. J. Grady II, [A. Siahkoohi](#), G. Rizzuti, P. A. Witte, O. Møyner, G. J. Gorman, and F. J. Herrmann. Learned multiphysics inversion with differentiable programming and machine learning. *The Leading Edge*, 42(7):474–486, 2023
[pdf] [link] [bib] [featured in Seismic Soundoff] [journal's most downloaded paper in '23]
- J4. Y. Zhang, Z. Yin, O. López, [A. Siahkoohi](#), M. Louboutin, R. Kumar, and F. J. Herrmann. Optimized time-lapse acquisition design via spectral gap ratio minimization. *Geophysics*, 88(4):A19–A23, 2023a
[pdf] [link] [bib]
- J3. [A. Siahkoohi](#), G. Rizzuti, R. Orozco, and F. J. Herrmann. Reliable amortized variational inference with physics-based latent distribution correction. *Geophysics*, 88(3):R297–R322, 2023a
[pdf] [slides] [code] [link] [bib] [featured in Geophysics Bright Spots]
- J2. [A. Siahkoohi](#), G. Rizzuti, and F. J. Herrmann. Deep Bayesian inference for seismic imaging with tasks. *Geophysics*, 87(5):S281–S302, 2022a
[pdf] [code] [link] [bib]
- J1. [A. Siahkoohi](#), M. Louboutin, and F. J. Herrmann. The importance of transfer learning in seismic modeling and imaging. *Geophysics*, 84(6):A47–A52, 2019a
[pdf] [code] [link] [bib]

Peer-Reviewed Conference Proceedings

- C35. [A. Siahkoohi](#) and H. Oh. Conditional neural control variates for variance reduction in Bayesian inverse problems. In *Proceedings of the Forty-second Conference on Uncertainty in Artificial Intelligence*, volume 337, pages 6321–6341, 2026
[pdf] [code] [poster] [link] [bib]
- C34. [A. Siahkoohi](#) and D. Sabeddu. On the role of memorization in learned priors for geophysical inverse problems. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, 2026. In print
[pdf] [code] [slides] [bib]

- C33. K. Aghazade, A. Siahkoohi, and A. Gholami. Scalable Bayesian full waveform inversion via dual augmented Lagrangian SVGD. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, 2026. In print
[pdf] [slides] [bib]
- C32. S. Alemohammad, J. Casco-Rodriguez, L. Luzi, A. I. Humayun, H. Babaei, D. LeJeune, A. Siahkoohi, and R. G. Baraniuk. Self-consuming generative models go MAD. In *The Twelfth International Conference on Learning Representations*, 2024
[pdf] [extended pdf] [poster] [link] [bib] [featured in the news 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]
- C31. L. Luzi, D. LeJeune, A. Siahkoohi, S. Alemohammad, V. Saragadam, H. Babaei, N. Liu, Z. Wang, and R. G. Baraniuk. Titan: Bringing the deep image prior to implicit representations. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 6165–6169, 2024b
[pdf] [code] [link] [bib]
- C30. L. Baldassari, A. Siahkoohi, J. Garnier, K. Sølna, and M. V. de Hoop. Conditional score-based diffusion models for Bayesian inference in infinite dimensions. In *Advances in Neural Information Processing Systems*, volume 36, pages 24262–24290, 2023
[pdf] [slides] [poster] [code] [link] [bib] [featured as a Spotlight presentation]
- C29. A. Siahkoohi, R. Morel, M. V. de Hoop, E. Allys, G. Sainton, and T. Kawamura. Unearthing InSights into Mars: Unsupervised source separation with limited data. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pages 31754–31772, 2023b
[pdf] [slides] [poster] [code] [link] [bib]
- C28. R. Orozco, M. Louboutin, A. Siahkoohi, G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Amortized normalizing flows for transcranial ultrasound with uncertainty quantification. In *Medical Imaging with Deep Learning*, volume 227, pages 332–349, 2023a
[pdf] [link] [bib]
- C27. R. Orozco, A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Refining amortized posterior approximations using gradient-based summary statistics. In *5th Symposium on Advances in Approximate Bayesian Inference*, 2023b
[pdf] [link] [bib]
- C26. R. Orozco, A. Siahkoohi, G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Adjoint operators enable fast and amortized machine learning based Bayesian uncertainty quantification. In *Medical Imaging 2023: Image Processing*, volume 12464, page 124641L, 2023c
[pdf] [link] [bib]
- C25. Y. Zhang, Z. Yin, O. Lopez, A. Siahkoohi, M. Louboutin, and F. J. Herrmann. 3D seismic survey design by maximizing the spectral gap. In *Third International Meeting for Applied Geoscience & Energy*, 2023b
[pdf] [poster] [bib]
- C24. A. Siahkoohi, M. Chinen, T. Denton, W. B. Kleijn, and J. Skoglund. Ultra-low-bitrate speech coding with pretrained Transformers. In *Proceedings of Interspeech*, pages 4421–4425, 2022b
[pdf] [link] [bib]
- C23. A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Velocity continuation with Fourier neural operators for accelerated uncertainty quantification. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1765–1769, 2022c
[pdf] [slides] [code] [link] [bib]
- C22. M. Louboutin, P. Witte, A. Siahkoohi, G. Rizzuti, Z. Yin, R. Orozco, and F. J. Herrmann. Accelerating innovation with software abstractions for scalable computational geophysics. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1482–1486, 2022
[pdf] [slides] [link] [bib]
- C21. Z. Yin, A. Siahkoohi, M. Louboutin, and F. J. Herrmann. Learned coupled inversion for carbon sequestration monitoring and forecasting with Fourier neural operators. In *Society of Exploration*

- Geophysicists Technical Program Expanded Abstracts*, pages 467–472, 2022
[pdf] [slides] [code] [link] [bib] [\[student oral paper honorable mention\]](#)
- C20. Y. Zhang, M. Louboutin, [A. Siahkoohi](#), Z. Yin, R. Kumar, and F. J. Herrmann. A simulation-free seismic survey design by maximizing the spectral gap. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 15–20, 2022
[pdf] [slides] [code] [link] [bib]
- C19. [A. Siahkoohi](#), R. Orozco, G. Rizzuti, and F. J. Herrmann. Wave-equation based inversion with amortized variational Bayesian inference. In *EAGE Deep learning for seismic processing: Investigating the foundations workshop*, 2022d
[pdf] [slides] [code] [link] [bib]
- C18. R. Orozco, [A. Siahkoohi](#), G. Rizzuti, T. van Leeuwen, and F. J. Herrmann. Photoacoustic imaging with conditional priors from normalizing flows. In *NeurIPS Workshop on Deep Learning and Inverse Problems*, 2021
[pdf] [poster] [link] [bib]
- C17. [A. Siahkoohi](#), G. Rizzuti, M. Louboutin, P. Witte, and F. J. Herrmann. Preconditioned training of normalizing flows for variational inference in inverse problems. In *3rd Symposium on Advances in Approximate Bayesian Inference*, 2021
[pdf] [slides] [code] [link] [bib]
- C16. [A. Siahkoohi](#) and F. J. Herrmann. Learning by example: Fast reliability-aware seismic imaging with normalizing flows. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1580–1585, 2021
[pdf] [slides] [code] [link] [bib]
- C15. R. Kumar, M. Kotsi, [A. Siahkoohi](#), and A. Malcolm. Enabling uncertainty quantification for seismic data preprocessing using normalizing flows (NF)—An interpolation example. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1515–1519, 2021
[pdf] [code] [link] [bib]
- C14. G. Rizzuti, [A. Siahkoohi](#), P. A. Witte, and F. J. Herrmann. Parameterizing uncertainty by deep invertible networks, an application to reservoir characterization. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1541–1545, 2020
[pdf] [slides] [code] [link] [bib]
- C13. M. Zhang, [A. Siahkoohi](#), and F. J. Herrmann. Transfer learning in large-scale ocean bottom seismic wavefield reconstruction. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1666–1670, 2020
[pdf] [slides] [code] [link] [bib]
- C12. [A. Siahkoohi](#), G. Rizzuti, and F. J. Herrmann. Weak deep priors for seismic imaging. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 2998–3002, 2020a
[pdf] [slides] [code] [link] [bib]
- C11. [A. Siahkoohi](#), G. Rizzuti, and F. J. Herrmann. Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 1636–1640, 2020b
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- C10. [A. Siahkoohi](#), G. Rizzuti, and F. J. Herrmann. A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification. In *European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts*, 2020c
[pdf] [slides] [code] [link] [bib]
- C9. F. J. Herrmann, [A. Siahkoohi](#), and G. Rizzuti. Learned imaging with constraints and uncertainty quantification. In *NeurIPS Deep Inverse Workshop*, 2019
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- C8. [A. Siahkoohi](#), R. Kumar, and F. J. Herrmann. Deep-learning based ocean bottom seismic wavefield recovery. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 2232–2237, 2019b
[pdf] [code] [slides] [link] [bib]
- C7. [A. Siahkoohi](#), D. J. Verschuur, and F. J. Herrmann. Surface-related multiple elimination with deep learning. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 4629–4634, 2019c
[pdf] [slides] [link] [bib]
- C6. G. Rizzuti, [A. Siahkoohi](#), and F. J. Herrmann. Learned iterative solvers for the Helmholtz equation. In *European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts*, 2019
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- C5. [A. Siahkoohi](#), M. Louboutin, R. Kumar, and F. J. Herrmann. Deep convolutional neural networks in prestack seismic—two exploratory examples. In *Society of Exploration Geophysicists Technical Program Expanded Abstracts*, pages 2196–2200, 2018a
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- C4. [A. Siahkoohi](#), R. Kumar, and F. J. Herrmann. Seismic data reconstruction with generative adversarial networks. In *European Association of Geoscientists & Engineers Conference and Exhibition Extended Abstracts*, 2018b
[pdf] [slides] [link] [bib]
- C3. [A. Siahkoohi](#) and A. Gholami. Sparsity promoting least squares migration for laterally inhomogeneous media. In *7th EAGE Saint Petersburg International Conference and Exhibition*, 2016
[pdf] [link] [bib]
- C2. M. S. Ebrahimi, M. H. Daraei, J. Rezaei, and [A. Siahkoohi](#). A novel utilization of wireless sensor networks as data acquisition system in smart grids. In *Materials Science and Information Technology*, volume 433-440, pages 6725–6730, 2012
[pdf] [link] [bib]
- C1. A. Najafi, [A. Siahkoohi](#), and M. B. Shamsollahi. A content-based digital image watermarking algorithm robust against JPEG compression. In *IEEE International Symposium on Signal Processing and Information Technology*, pages 432–437, 2011
[pdf] [link] [bib]

Theses

- T1. [A. Siahkoohi](#). *Deep generative models for solving geophysical inverse problems*. PhD thesis, **Georgia Institute of Technology**, 2022
[pdf] [slides] [link] [bib]

Technical Reports

- R4. P. M. Mayer, L. Luzi, [A. Siahkoohi](#), D. H. Johnson, and R. G. Baraniuk. Improving fairness and mitigating MADness in generative models. Technical Report arXiv:2405.13977, Rice University, 2024
[pdf] [code] [slides] [bib]
- R3. L. Baldassari, [A. Siahkoohi](#), J. Garnier, K. Sølna, and M. V. de Hoop. Taming score-based diffusion priors for infinite-dimensional nonlinear inverse problems. Technical Report arXiv:2405.15676, Rice University, 2024
[pdf] [bib]
- R2. [A. Siahkoohi](#), G. Rizzuti, P. A. Witte, and F. J. Herrmann. Faster uncertainty quantification for inverse problems with conditional normalizing flows. Technical Report arXiv:2007.07985, Georgia Institute of Technology, 2020d [pdf] [link] [bib]
- R1. [A. Siahkoohi](#), M. Louboutin, and F. J. Herrmann. Neural network augmented wave-equation simulation.

Technical Report arXiv:1910.00925, Georgia Institute of Technology, 2019d
[pdf] [code] [link] [bib]

Awards

Future Faculty Fellows Award

Rice University, George R. Brown School of Engineering and Computing
[link]

June 2024
Houston, TX, USA

Mentoring Experience

University of Central Florida

Departments of Computer Science & Electrical and Computer Engineering

Orlando, FL, USA

- ▶ Banafsheh Adami [link] 2026 – Present
CS Ph.D. Student — co-advised with Alain Kassab
- ▶ Anirudh Thatipelli [link] 2025 – Present
CS Ph.D. Student — Ph.D. advisor
- ▶ Davide Sabeddu [link] 2025 – 2026
ECE Ph.D. Student — Advised on the development of methods for mitigating overfitting in generative models and co-authored a paper (A. Siahkoohi and Sabeddu, 2026)

Rice University

Department of Computational Applied Mathematics & Operations Research

Houston, TX, USA

- ▶ Jeffrey J. Sam [link] 2024 – 2025
M.Sc. Student — Advised on the design and implementation of experiments and co-authored a paper (Thatipelli et al., 2026)
- ▶ Paul M. Mayer [link] 2022 – 2025
Ph.D. Student — Advised on the development of methods and software for two projects and co-authored two papers (Luzi et al., 2024a; Mayer et al., 2024)

Georgia Institute of Technology

School of Computational Science and Engineering

Atlanta, GA, USA

- ▶ Rafael Orozco [link] 2020 – 2022
Ph.D. Student — Advised on the development of methods and software for main Ph.D. thesis and co-authored four papers (Orozco et al., 2021, 2023b,c, 2025)
- ▶ Mi Zhang [link] 2019 – 2020
Visiting Ph.D. Student (China University of Petroleum-Beijing) — Advised on the development of methods and software for a project and co-authored a paper (Zhang et al., 2020)

Teaching Experience

University of Central Florida

Department of Computer Science

Orlando, FL, USA

- ▶ Algorithms for Machine Learning [link] Fall 2026
Instructor of Record
- ▶ Sampling Methods for Uncertainty Quantification [link] Spring 2026
Instructor of Record
- ▶ Algorithms for Machine Learning [link] Fall 2025
Instructor of Record

Rice University

Department of Computational Applied Mathematics & Operations Research

Houston, TX, USA

- ▶ Numerical Analysis Fall 2024

Substitute Instructor (12 lectures)	
▶ Numerical Analysis I	Fall 2022
Substitute Instructor (18 lectures)	
Georgia Institute of Technology	Atlanta, GA, USA
School of Computational Science and Engineering	
▶ Computational Foundations of Machine Learning	Spring 2022
Teaching Assistant	
▶ Imaging with Data-Driven Models	Fall 2019
Teaching Assistant	
▶ Numerical Analysis I	Fall 2018
Teaching Assistant	
Sharif University of Technology	Tehran, Iran
Department of Electrical Engineering	
▶ Digital Signal Processing	Spring 2011
Teaching Assistant	
▶ Signals and Systems	Spring 2011
Teaching Assistant	
▶ Linear Algebra	Spring 2010
Teaching Assistant	
▶ Electrical Engineering: Principles and Laboratory	Fall 2009
Teaching Assistant	

Talks

Invited Talks

T35. University of Central Florida	April 2026
On the role of memorization in learned priors for geophysical inverse problems	Oral presentation
Foundations of AI Seminar	
T34. IEEE Computer Society, Chapter of San Diego	June 2025
Mitigating biases in self-consuming generative models	Virtual oral presentation
Open Research Institute (ORI)	
[video]	
T33. University of Central Florida	April 2025
Towards reliable AI: A framework for quantification of AI uncertainty	Oral presentation
Department of Computer Science	
T32. Montana State University	March 2025
Towards reliable AI: A framework for quantification of AI uncertainty	Oral presentation
Gianforte School of Computing	
T31. The University of California, Santa Barbara	February 2025
Towards reliable AI: A framework for quantification of AI uncertainty	Oral presentation
Department of Mechanical Engineering	
T30. Johns Hopkins University	January 2025
Towards reliable AI: A framework for quantification of AI uncertainty	Oral presentation
Department of Electrical and Computer Engineering	
T29. ISCL Seminar Series, Pennsylvania State University	November 2024
Mitigating biases in self-consuming generative models	Virtual oral presentation
Interdisciplinary Scientific Computing Laboratory (Dr. Romit Maulik)	
[video]	

- T28. **CNRS, Université Montpellier** January 2023
Low-cost uncertainty quantification for large-scale inverse problems Virtual oral presentation
RhEoVOLUTION Group (Dr. Andréa Tommasi)
- T27. **Workshop on Subsurface Uncertainty Description and Estimation** August 2022
Reliable amortized variational inference with conditional normalizing flows via physics-based latent distribution correction Oral presentation
International Meeting for Applied Geoscience & Energy
- T26. **Intelligent illumination of the Earth Workshop** June 2021
Fast and reliability-aware seismic imaging with conditional normalizing flows Virtual oral presentation
King Abdullah University of Science and Technology
- T25. **Advances in Seismic Imaging and Inversion Mini-symposium** October 2020
Unsupervised data-guided uncertainty analysis in imaging and horizon tracking Virtual oral presentation
The 3rd Annual Meeting of the SIAM Texas–Louisiana Section

Contributed Talks

- T24. **International Meeting for Applied Geoscience & Energy** August 2026
On the role of memorization in learned priors for geophysical inverse problems Oral presentation
- T23. **International Meeting for Applied Geoscience & Energy** August 2026
Scalable Bayesian full waveform inversion via dual augmented Lagrangian SVGD Oral presentation
- T22. **Conference on Uncertainty in Artificial Intelligence** August 2026
Conditional neural control variates for variance reduction in Bayesian inverse problems Poster presentation
- T21. **SIAM Annual Meeting** July 2026
Conditional neural control variates for variance reduction in Bayesian inverse problems Poster presentation
- T20. **UCF Spring AI Research Day, UCF Institute of Artificial Intelligence** February 2026
Multi-scale clustering and source separation of InSight mission seismic data Oral presentation
- T19. **Geo-Mathematical Imaging Group Partners Meeting, Rice University** November 2024
Improving fairness and mitigating MADness in generative models Oral presentation
- T18. **International Conference on Machine Learning** July 2023
Unearthing InSights into Mars: Unsupervised source separation with limited data Poster presentation
- T17. **Symposium on Advances in Approximate Bayesian Inference** July 2023
Refining amortized posterior approximations using gradient-based summary statistics Poster presentation
- T16. **Geo-Mathematical Imaging Group Partners Meeting, Rice University** May 2023
Martian time-series unraveled: A multi-scale nested approach with factorial variational autoencoders Oral presentation
- T15. **Geo-Mathematical Imaging Group Partners Meeting, Rice University** May 2023
Unearthing InSights into Mars: Unsupervised source separation with limited data Oral presentation
- T14. **International Meeting for Applied Geoscience & Energy** August 2022
Velocity continuation with Fourier neural operators for accelerated uncertainty quantification Oral presentation
- T13. **Chrome Media Team, Google** December 2021
Low-bitrate speech coding with Transformers Virtual oral presentation
- T12. **ML4SEISMIC Partners Meeting, Georgia Institute of Technology** November 2021
Multifidelity conditional normalizing flows for physics-guided Bayesian Virtual oral presentation

inference

- T11. **ML4SEISMIC Partners Meeting, Georgia Institute of Technology** November 2021
Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach Virtual oral presentation
- T10. **Society of Exploration Geophysicists International Exposition and Annual Meeting** September 2021
Learning by example: Fast reliability-aware seismic imaging with normalizing flows Virtual oral presentation
[video]
- T9. **Symposium on Advances in Approximate Bayesian Inference** January 2021
Preconditioned training of normalizing flows for variational inference in inverse problems Prerecorded short oral presentation
[video]
- T8. **European Association of Geoscientists & Engineers Annual Conference & Exhibition** December 2020
A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification Virtual oral presentation
- T7. **Society of Exploration Geophysicists International Exposition and Annual Meeting** October 2020
Uncertainty quantification in imaging and automatic horizon tracking—A Bayesian deep-prior based approach Virtual oral presentation
[video]
- T6. **Society of Exploration Geophysicists International Exposition and Annual Meeting** October 2020
Weak deep priors for seismic imaging Virtual oral presentation
[video]
- T5. **Society of Exploration Geophysicists Student Chapter, Georgia Tech** February 2020
A deep-learning based Bayesian approach to seismic imaging and uncertainty quantification Oral presentation
- T4. **HotCSE Seminar, CSE Department, Georgia Institute of Technology** November 2019
Learned imaging with constraints and uncertainty quantification Oral presentation
- T3. **Society of Exploration Geophysicists International Exposition & Annual Meeting** September 2019
Deep-learning based ocean bottom seismic wavefield recovery Oral presentation
- T2. **Society of Exploration Geophysicists International Exposition & Annual Meeting** September 2019
Surface-related multiple elimination with deep learning Oral presentation
- T1. **Society of Exploration Geophysicists International Exposition & Annual Meeting** October 2018
Deep convolutional neural networks in prestack seismic—two exploratory examples Poster presentation

Professional Service

Editorial Service

- **Acta Geophysica**, Associate Editor 2024 – Present
Applied Geophysics section
- **Journal of Mathematics**, Guest Editor 2023 – 2024
Special issue on Applied Mathematics in Inverse Problems and Uncertainty Quantification

Conference Organization

- **International Meeting for Applied Geoscience & Energy**, Session Chair 2022

Technical Program Committee Member and Reviewer

- ▶ The Conference on Uncertainty in Artificial Intelligence (UAI) 2026
- ▶ Symposium on Probabilistic Machine Learning (ProbML) 2026
- ▶ International Conference on Machine Learning (ICML) 2024 – 2026
- ▶ International Conference on Learning Representations (ICLR) 2024 – 2026
- ▶ Annual AAAI Conference on Artificial Intelligence 2025 – 2026
- ▶ Structured Probabilistic Inference & Generative Modeling 2023 – 2025
- ▶ Frontiers in Probabilistic Inference: Sampling Meets Learning 2025
- ▶ Neural Information Processing Systems (NeurIPS) 2023 – 2025
- ▶ Artificial Intelligence and Statistics Conference (AISTATS) 2024 – 2025
- ▶ Advances in Approximate Bayesian Inference (AABI) 2023 – 2024
- ▶ Structured Probabilistic Inference & Generative Modeling (ICML workshop) 2023 – 2024
- ▶ International Speech Communication Association (Interspeech) 2023
- ▶ Deep Generative Models for Health (NeurIPS workshop) 2023
- ▶ International Meeting for Applied Geoscience & Energy 2023

Journal Reviewer

- ▶ Transactions on Machine Learning Research
- ▶ IEEE Transactions on Neural Networks and Learning Systems
- ▶ IEEE Transactions on Computational Imaging
- ▶ Journal of Medical Imaging
- ▶ SIAM Journal on Scientific Computing
- ▶ Notices of the American Mathematical Society (AMS)
- ▶ Entropy
- ▶ IEEE Transactions on Geoscience and Remote Sensing
- ▶ IEEE Geoscience and Remote Sensing Letters
- ▶ Remote Sensing
- ▶ Geosciences
- ▶ Geophysics
- ▶ The Leading Edge
- ▶ Geophysical Prospecting
- ▶ Journal of Geophysical Research – Solid Earth

Industry Research Experience

Google

Research Intern (cf. [A. Siahkoohi et al. \(2022b\)](#))
Chrome Media Team

August 2021 – December 2021
San Francisco, CA, USA

Selected Media Coverage

Future Faculty Fellow Ali Siahkoohi joins University of Central Florida as assistant professor June 2025
Rice University Engineering News
[\[link\]](#)

AI's Mad Loops
Rice Magazine
[\[link\]](#)

February 2025

AI Appears to Be Slowly Killing Itself Futurism [link]	August 2024
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